Java Assignment

28 Programs

1. Hello World!

```
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

2. Prime Number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        boolean prime = n > 1;
        for (int i = 2; i * i <= n; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }
        if (prime) {
            System.out.println(n + " is a prime number");
        } else {
            System.out.println(n + " is not a prime number");
        }
    }
}
```

3. Palindrome Number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int temp = n;
        int reverse = 0;
        while (temp > 0) {
            reverse = reverse * 10 + temp % 10;
            temp = temp / 10;
        }
        if (n == reverse) {
            System.out.println(n + " is a palindrome number");
        } else {
            System.out.println(n + " is not a palindrome number");
        }
    }
}
```

4. Armstrong Number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int digits = String.valueOf(n).length();
        int temp = n;
        int sum = 0;
        while (temp > 0) {
            int d = temp % 10;
            sum = sum + (int) Math.pow(d, digits);
            temp = temp / 10;
        }
        if (sum == n) {
            System.out.println(n + " is an Armstrong number");
        } else {
            System.out.println(n + " is not an Armstrong number");
        }
    }
}
```

5. Strong Number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int temp = n;
        int sum = 0;
        while (temp > 0) {
            int d = temp % 10;
            int fact = 1;
            for (int i = 1; i <= d; i++) {
                fact = fact * i;
            }
            sum = sum + fact;
            temp = temp / 10;
        }
        if (sum == n) {
            System.out.println(n + " is a strong number");
        } else {
            System.out.println(n + " is not a strong number");
        }
    }
}
```

6. Even - Odd

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        if (n % 2 == 0) {
            System.out.println(n + " is even");
        } else {
            System.out.println(n + " is odd");
        }
    }
}
```

7. Leap Year

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a year: ");
        int year = sc.nextInt();
        if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0) {
            System.out.println(year + " is a leap year");
        } else {
            System.out.println(year + " is not a leap year");
        }
    }
}
```

8. Celsius to Fahrenheit

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();
        double fahrenheit = (celsius * 9 / 5) + 32;
        System.out.println("Temperature in Fahrenheit: " + fahrenheit);
    }
}
```

9. Perfect Number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int sum = 0;
        for (int i = 1; i < n; i++) {
            if (n % i == 0) {
                sum = sum + i;
            }
        }
        if (sum == n && n > 0) {
            System.out.println(n + " is a perfect number");
        } else {
            System.out.println(n + " is not a perfect number");
        }
    }
}
```

10. Fibonacci Series

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of terms: ");
        int n = sc.nextInt();
        int a = 0;
        int b = 1;
        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
        }
        System.out.println();
    }
}
```

11. Factorial

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        System.out.println("Factorial of " + n + " is " + fact);
    }
}
```

12. Sum of Digit

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int sum = 0;
        while (n > 0) {
            sum = sum + n % 10;
            n = n / 10;
        }
        System.out.println("Sum of digits: " + sum);
    }
}
```

13. Star Triangle

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int n = sc.nextInt();
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

14. Pyramid

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int n = sc.nextInt();
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= 2 * i - 1; k++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

15. Diamond

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int n = sc.nextInt();
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= 2 * i - 1; k++) {
                System.out.print("*");
            }
            System.out.println();
        }
        for (int i = n - 1; i >= 1; i--) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= 2 * i - 1; k++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

16. Number Swap

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        System.out.println("Before swap: a = " + a + ", b = " + b);
        int temp = a;
        a = b;
        b = temp;
        System.out.println("After swap: a = " + a + ", b = " + b);
    }
}
```

17. Decimal to Binary

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a decimal number: ");
        int n = sc.nextInt();
        if (n == 0) {
            System.out.println("Binary: 0");
            return;
        }
        String binary = "";
        while (n > 0) {
            binary = (n % 2) + binary;
            n = n / 2;
        }
        System.out.println("Binary: " + binary);
    }
}
```

18. Simple Calculator (Switch-Case)

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        double a = sc.nextDouble();
        System.out.print("Enter second number: ");
        double b = sc.nextDouble();
        System.out.print("Enter operator (+, -, *, /): ");
        char op = sc.next().charAt(0);
        switch (op) {
            case '+':
                System.out.println("Result: " + (a + b));
                break;
            case '-':
                System.out.println("Result: " + (a - b));
                break;
            case '*':
                System.out.println("Result: " + (a * b));
                break;
            case '/':
                if (b != 0) {
                    System.out.println("Result: " + (a / b));
                } else {
                    System.out.println("Cannot divide by zero");
                }
                break;
            default:
                System.out.println("Invalid operator");
        }
    }
}
```

19. Largest among 3 Numbers

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        System.out.print("Enter third number: ");
        int c = sc.nextInt();
        int largest = a;
        if (b > largest) {
            largest = b;
        }
        if (c > largest) {
            largest = c;
        }
        System.out.println("Largest number: " + largest);
    }
}
```

20. Series: 1 + 2/4 + 3/8 + 4/16 + ... + n/2^n

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter n: ");
        int n = sc.nextInt();
        double sum = 1;
        for (int i = 2; i <= n; i++) {
            sum = sum + (double) i / Math.pow(2, i);
        }
        System.out.println("Sum of the series: " + sum);
    }
}
```

21. Password (do-while)

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String correctPassword = "admin123";
        String input;
        do {
            System.out.print("Enter password: ");
            input = sc.nextLine();
            if (!input.equals(correctPassword)) {
                System.out.println("Wrong password. Try again.");
            }
        } while (!input.equals(correctPassword));
        System.out.println("Login successful");
    }
}
```

22. Student Class with Object

```
public class Main {
    public static void main(String[] args) {
        Student s1 = new Student(1, "Rahim", 85.5);
        Student s2 = new Student(2, "Karim", 90.0);
        s1.display();
        System.out.println();
        s2.display();
    }
}

class Student {
    int id;
    String name;
    double marks;

    Student(int id, String name, double marks) {
        this.id = id;
        this.name = name;
        this.marks = marks;
    }

    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
}
```

23. Use of Local and Instance Variable

```
public class Main {
    int instanceVariable = 10;

    void show() {
        int localVariable = 20;
        System.out.println("Instance variable: " + instanceVariable);
        System.out.println("Local variable: " + localVariable);
    }

    public static void main(String[] args) {
        Main obj = new Main();
        obj.show();
    }
}
```

24. Use of Shift Operator

```
public class Main {
    public static void main(String[] args) {
        int a = 8;
        int b = -8;
        System.out.println("a << 2 = " + (a << 2));
        System.out.println("a >> 2 = " + (a >> 2));
        System.out.println("b >> 2 = " + (b >> 2));
        System.out.println("b >>> 28 = " + (b >>> 28));
    }
}
```

25. Leap Year (Ternary Operator)

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a year: ");
        int year = sc.nextInt();
        String result = ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
                ? "Leap year" : "Not a leap year";
        System.out.println(year + " is: " + result);
    }
}
```

26. GCD

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        System.out.println("GCD: " + a);
    }
}
```

27. LCM

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        int x = a;
        int y = b;
        while (y != 0) {
            int temp = y;
            y = x % y;
            x = temp;
        }
        int lcm = (a / x) * b;
        System.out.println("LCM: " + lcm);
    }
}
```

28. Print How You Felt After Doing This

```
public class Main {
    public static void main(String[] args) {
        System.out.println("I felt very happy and proud after completing this assignment!");
    }
}
```